Code No: 135AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
DATA COMMUNICATION AND COMPUTER NETWORKS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	List out the topologies used in networks.	[2]
b)	Define Multiplexing. Write about FDMA.	[3]
c)	Explain flow control.	[2]
d)	List various ALOHA protocols.	[3]
e)	Describe why the class C is most commonly used Network class.	[2]
f)	What is fragmentation? Write its significance.	[3]
g)	What is a socket in transport layer?	[2]
h)	Why three way handshake is used in TCP.	[3]
i)	What is the header format of HTTP reply message?	[2]
j)	What are the key components in SMTP protocol?	[3]

PART – B

(50 Marks)

2.a)	List and explain about various layers in TCP/IP model.	
b)	What are the steps involved in virtual circuit setup?	[5+5]
	OR	
3.a)	Briefly explain about frame relay networks.	
b)	Explain in detail about various types of multiplexing.	[5+5]
4.a)	What is the role of CSMA in multiple access? Explain.	
b)	What is hamming code? Explain its purpose with an example.	[5+5]
	OR	
5.a)	Compare and contrast the Go-Back-N ARQ Protocol with Selective-Repeat ARQ.	
b)	Explain about various framing methods.	[5+5]
6.a)	Explain the shortest path routing with an example.	
b)	List and explain in detail about the applications of tunneling.	[5+5]
	OR	
7.a)	Explain the significance of options field of IP packet.	
b)	Differentiate between ARP and RARP with an examples.	[5+5]

8.a) Discuss about the TCP sliding window protocol for error control.
b) Briefly explain about TCP deals with congestion control. [5+5]

OR

9.a) Explain the role of quality of service in Internet.
b) With a neat diagram, explain the TCP header fields in detail. [5+5]

10.a) Illustrate the role of POP3 in electronic mail applications.
b) Explain the role of domain name service in Internet. [5+5]

OR

11.a) Explain how a file transfer is done using FTP?
b) Describe in detail about the concept of Telnet. [5+5]

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